

React 19 & React Native WebTypeScriptSupabase PostgreSQL & RLSDijkstra Shortest PathPoint-in-Polygon Raycasting

2D/3D Isometric Vector Map1-Click Guest Access

⚠️ Problem Statement

University students face attendance debarment due to lack of real-time visibility and predictive forecasting. Concurrently, large multi-block university campuses cause disorientation, with students and visitors struggling to locate departmental rooms, laboratories, and faculty offices.

💡 Engineered Solution

A unified, cross-platform academic operating system providing instant mathematical bunk calculation, dynamic timetable scheduling, and an in-browser vector GIS map with multi-floor room lookup and perimeter-respecting shortest path navigation.

KEY SYSTEM INNOVATIONS

📊 1. Predictive Bunk Simulator

- **Master Circular Dial:** Real-time compliance tracking against institutional 75% thresholds.
- **In-Memory Math Sandbox:** Calculates exact *Safe Bunks* remaining and *Recovery Classes* needed without altering database records.
- **Live Lecture Stopwatch:** Monitors ongoing class durations with active status badges.

🗺️ 2. 2D/3D Campus GIS Navigation

- **Isometric 3D Extrusion:** Custom SVG rendering of 23 campus buildings with depth extrusion and shadows.
- **Deterministic Raycasting:** Point-in-polygon math eliminates touch errors; strictly selects true building contours.
- **Dijkstra Wayfinding:** 87 nodes & 95 edges routing around building walls with live road-snapped GPS tracking.

📅 3. Dynamic Timetable & Deep Linking

- **Weekly Schedule:** Day-by-day class timeline with room numbers, faculty tags, and subject codes.
- **1-Tap Map Launcher:** Clicking the map icon next to any upcoming lecture auto-focuses the target building and switches to that exact floor.
- **Multi-Floor Indoor Directory:** Searchable database of engineering labs, HOD offices, and lecture halls.

🔧 4. Standalone Route Studio Pro

- **In-Browser CAD/GIS Tool:** (CampusRouteStudio.html) Standalone suite for editing campus walkways.
- **Magnetic Waypoint Snapping:** Nodes snap within 12px to stitch intersections and crosswalks automatically.
- **Distance Ruler & Export:** Measures physical meters on campus; exports in JSON & TypeScript formats.

TECHNICAL ARCHITECTURE & MATHEMATICAL MODELS

Subsystem	Technologies & Libraries	Key Algorithms & Mathematical Models
Frontend Web / PWA	React 19, React Native Web, Expo SDK 57, TypeScript	Component-level memoization, adaptive safe-area insets
Vector Graphics	React Native SVG, Reanimated, Lucide Vector Icons	Isometric 3D projection, dynamic animated polyline marching
Wayfinding & Spatial	Custom Graph Adjacency List, Coordinate Engine	Dijkstra shortest path algorithm, Raycasting (Jordan Curve Theorem)
GPS Tracking	Expo Location API, Affine Transform Matrix	Low-pass EMA smoothing ($\alpha=0.35$), orthogonal road centerline projection
Backend & Database	Supabase, PostgreSQL 15, Vercel Edge Hosting	Row-Level Security (RLS), Realtime WebSocket Postgres channels

🏆 Ready for Evaluation — Instant 1-Click Access

Open <https://attendancetracker26.vercel.app/> and click "Guest Mode" in the top-right header to instantly test with full pre-loaded university data.

Local Run: `npx serve dist`
Dev Mode: `npm run web`